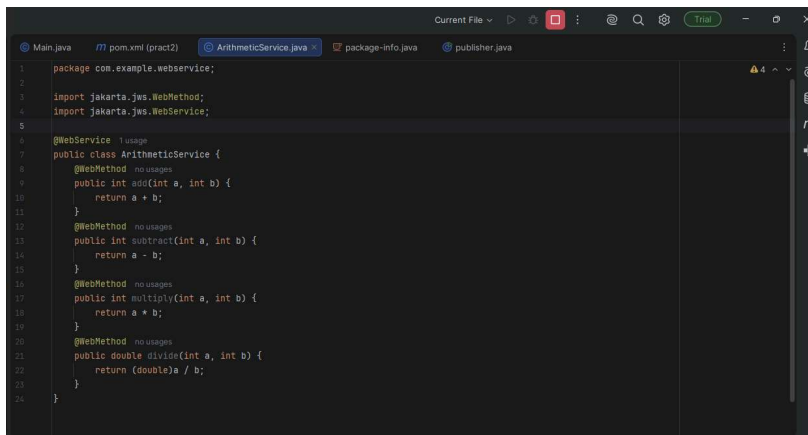
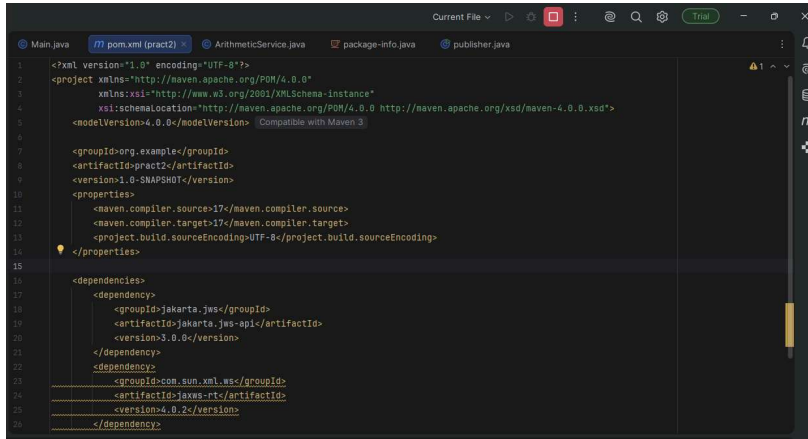
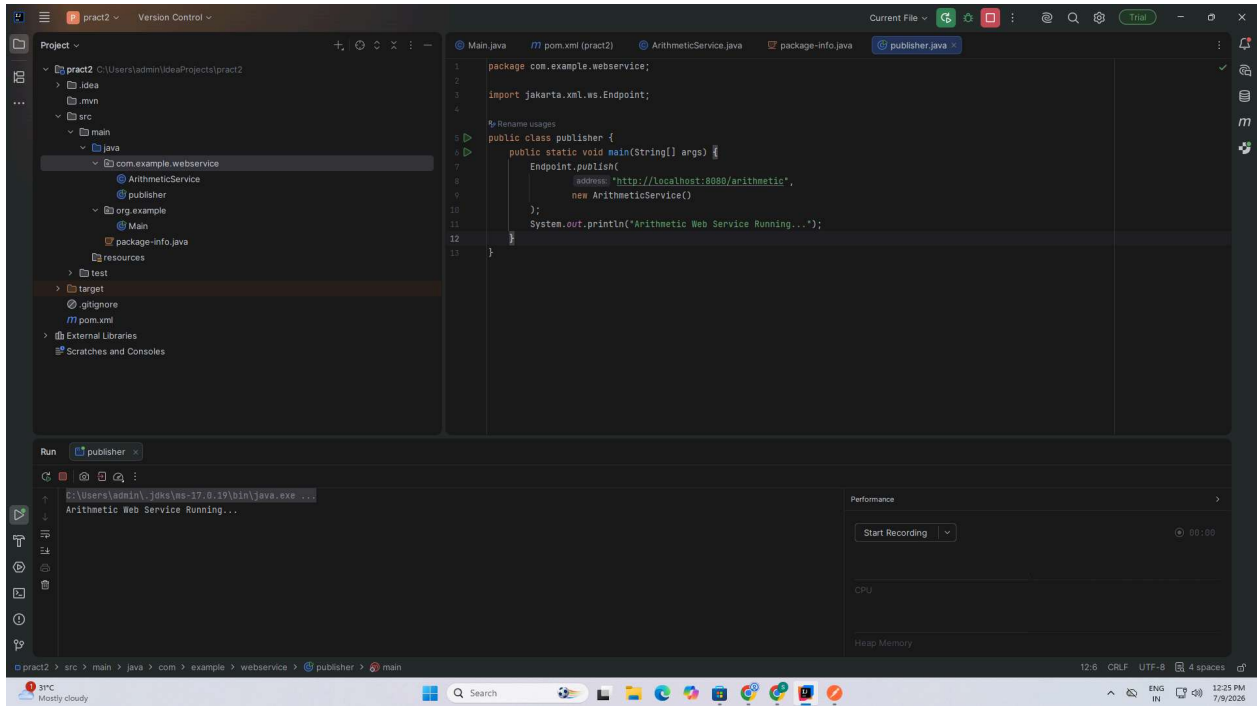
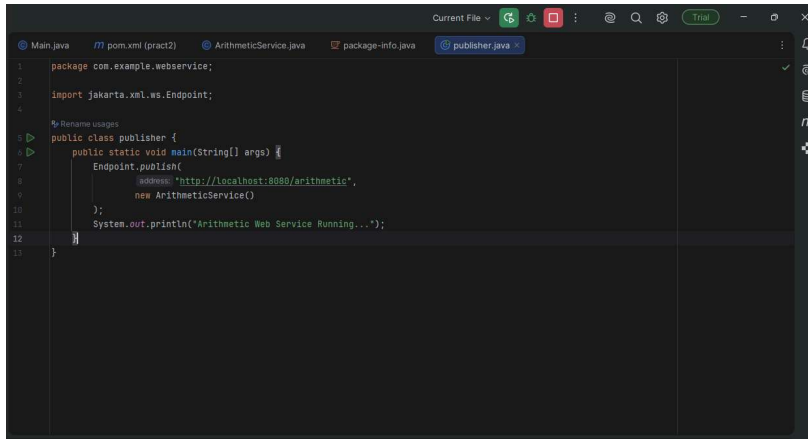


Name: Sujay Sunil Mane
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Batch - 3
Subject: Web Services

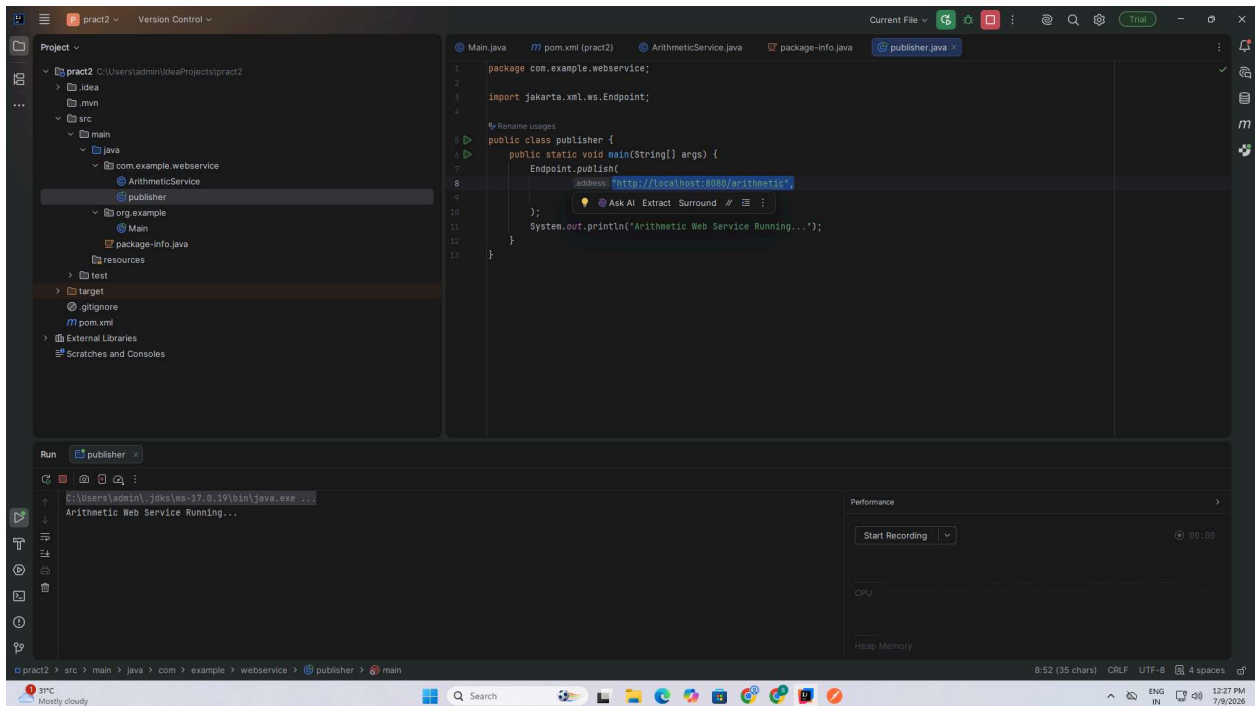
Practical - 2

- 1)Open **IntelliJ IDEA**.
- 2)Create a new **Spring Boot Project** named **web-service**.
- 3)Add the required dependencies:
 - Spring Web
 - Spring Web Services
 - Spring Boot DevTools
- 4)Click **Create** to generate the project.
- 5)Open the project in IntelliJ IDEA.
- 6)Create the required packages and Java classes:
 - `Application.java`
 - `WebServiceConfig.java`
 - `CountriesEndpoint.java`
- 7)Create the **countries.xsd** file inside the **src/main/resources** folder.
- 8)Configure the SOAP Web Service in **WebServiceConfig.java**.
- 9)Implement the business logic in **CountriesEndpoint.java** using the `@Endpoint` annotation.
- 10)Build the project using **Maven**.
- 11)Run the application by executing **Application.java**.
- 12)Wait until the application starts successfully on **localhost:8080**.





```
1 package com.example.webservice;
2
3 import jakarta.xml.ws.Endpoint;
4
5 // Rename usages
6 public class Publisher {
7     public static void main(String[] args) {
8         Endpoint.publish(
9             new ArithmeticService(),
10             "http://localhost:8080/arithmetic");
11     }
12     System.out.println("Arithmetic Web Service Running...");
13 }
```



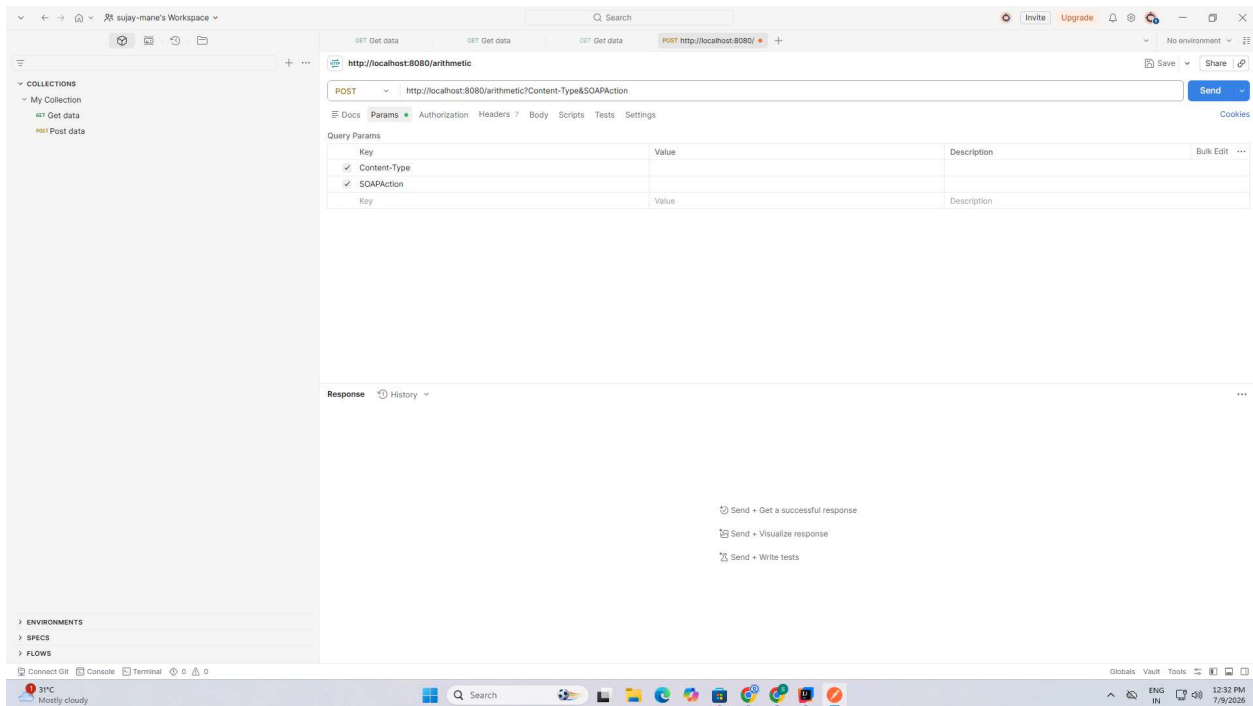
13)Open a web browser and verify the WSDL by visiting:

<http://localhost:8080/ws/countries.wsdl>



14)Open **Postman**.

15)Create a new **POST** request.

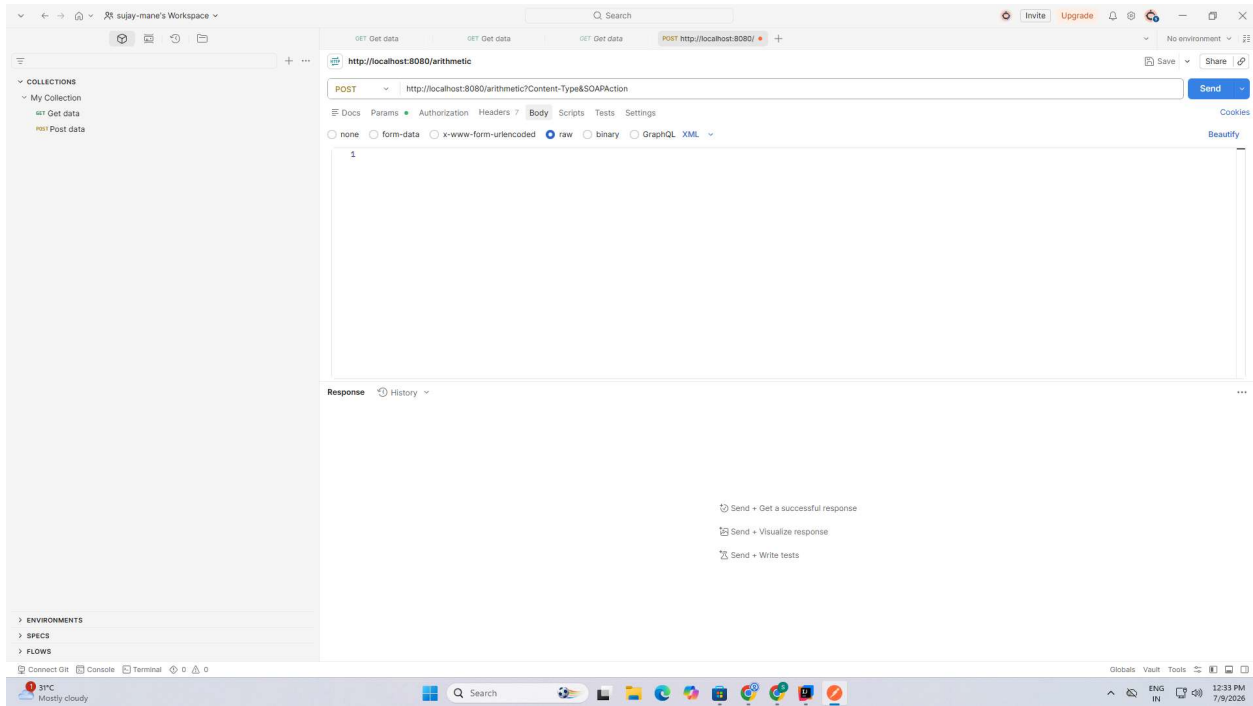


16)Enter the URL:

http://localhost:8080/ws

17)In the **Headers** tab, add:

Content-Type : text/xml

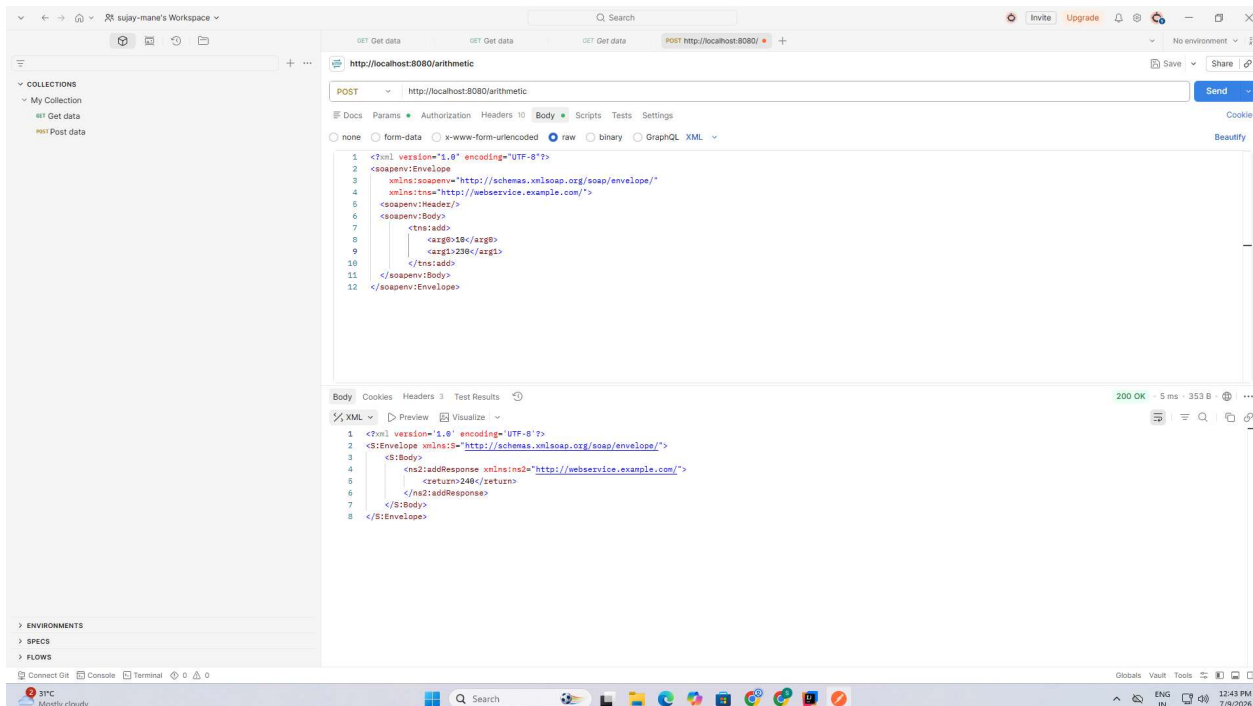


18) Select **Body** → **Raw** → **XML**.

19) Paste the SOAP XML request.

20) Click **Send**

Addition:



Verify that the SOAP XML response is received successfully.

Subtraction :

The screenshot shows a Postman interface with a workspace named "sujay-mane's Workspace". The active collection is "My Collection", and the selected item is "Post data". The environment is set to "No environment". The request is a POST to "http://localhost:8080/arithmetic". The request body is a SOAP message:

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <soapenv:Envelope
3   xmlns:soapenv='http://schemas.xmlsoap.org/soap/envelope/'
4   xmlns:tns='http://webservice.example.com/'>
5   <soapenv:Header/>
6   <soapenv:Body>
7     <tns:subtract>
8       <arg0>10</arg0>
9       <arg1>200</arg1>
10    </tns:subtract>
11  </soapenv:Body>
12 </soapenv:Envelope>
```

The response is a 200 OK status with a SOAP message:

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <S:Envelope xmlns:S='http://schemas.xmlsoap.org/soap/envelope/'>
3   <S:Body>
4     <ns2:subtractResponse xmlns:ns2='http://webservice.example.com/'>
5       <return>-228</return>
6     </ns2:subtractResponse>
7   </S:Body>
8 </S:Envelope>
```

The status bar at the bottom shows "200 OK", "5 ms", and "364 B".

Multiplication:

The screenshot shows a Postman interface with a workspace named "sujay-mane's Workspace". The active collection is "My Collection", and the selected item is "Post data". The environment is set to "No environment". The request is a POST to "http://localhost:8080/arithmetic". The request body is a SOAP message:

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <soapenv:Envelope
3   xmlns:soapenv='http://schemas.xmlsoap.org/soap/envelope/'
4   xmlns:tns='http://webservice.example.com/'>
5   <soapenv:Header/>
6   <soapenv:Body>
7     <tns:multiply>
8       <arg0>10</arg0>
9       <arg1>228</arg1>
10    </tns:multiply>
11  </soapenv:Body>
12 </soapenv:Envelope>
```

The response is a 200 OK status with a SOAP message:

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <S:Envelope xmlns:S='http://schemas.xmlsoap.org/soap/envelope/'>
3   <S:Body>
4     <ns2:multiplyResponse xmlns:ns2='http://webservice.example.com/'>
5       <return>2280</return>
6     </ns2:multiplyResponse>
7   </S:Body>
8 </S:Envelope>
```

The status bar at the bottom shows "200 OK", "6 ms", and "364 B".

Division:

The screenshot displays a REST client interface with a sidebar on the left containing 'COLLECTIONS' (My Collection, GET Get data, POST Post data), 'ENVIRONMENTS', 'SPECS', and 'FLOWS'. The main panel shows a POST request to 'http://localhost:8080/arithmetic'. The request body is a SOAP envelope for a division operation. The response is a 200 OK status with a SOAP envelope containing the result of the division.

Request:

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <soapenv:Envelope
3   xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
4   xmlns:tns="http://webservice.example.com/"
5 <soapenv:Header/>
6 <soapenv:Body>
7   <tns:divide>
8     <arg0>10</arg0>
9     <arg1>300</arg1>
10  </tns:divide>
11 </soapenv:Body>
12 </soapenv:Envelope>
```

Response:

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
3 <S:Body>
4   <ns2:divideResponse xmlns:ns2="http://webservice.example.com/"
5     <return>0.043478260869565216</return>
6   </ns2:divideResponse>
7 </S:Body>
8 </S:Envelope>
```

The bottom status bar shows system information: 31°C, Mostly cloudy, and the date/time 12:48 PM 7/9/2026.